

FULL-GRID DIFFERENTIABLE ROUTING

Patch-8 stem
784 + CLS

Local encoder $\times 6$
28 $\times$ 28 grid

Spatial mass routing $\times 6$
radius $R = 3$

soft transport keeps all slots

full-grid features
for recovery K/V

first shorter tensor

PHYSICALLY COMPRESSED GLOBAL REASONING

CLS head
ImageNet

Latent encoder $\times 6$
392 + CLS $\cdot \log m$

Recovery cross-attn
392 Q $\cdot$ 784 K/V

Exact-budget gather
784 $\rightarrow$ 392

Spatial support restricts admissible routes; similarities, masses, and carrier locations remain learned.